Experiment – 8: Laplace Transform, Inverse Transform, ROC and Stability.

OBJECTIVES:

ii) To determine the inverse Laplace transform of a given frequency-domain representation and reconstruct the corresponding time-domain signal.

iii) To determine the Region of Convergence (ROC) of Laplace transforms and analyze its significance in characterizing continuous-time signals and systems**.**

(ii) Determination of Inverse Laplace Transform for Signal Reconstruction.

Software / Laboratory Environment:

MATLAB

MATLAB Coding:

```
clc;
clear;
close all;

% Experiment 8(ii): Inverse Laplace Transform

syms s t

X = 1/((s+2)*(s+3));

% Calculate inverse Laplace transform
x = ilaplace(X,s,t);

disp('Inverse Laplace Transform x(t) =');
pretty(x);
```

```
% Reconstruct the signal
tt = 0:0.001:5;
xx = double(subs(x,t,tt));

% Plot the reconstructed signal
figure;
plot(tt,xx,'LineWidth',1.2);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Signal Reconstructed using Inverse Laplace Transform');

clc;
clear;
close all;

% Experiment 8(ii): Inverse Laplace Transform

syms s t

X = 1/((s+2)*(s+3));

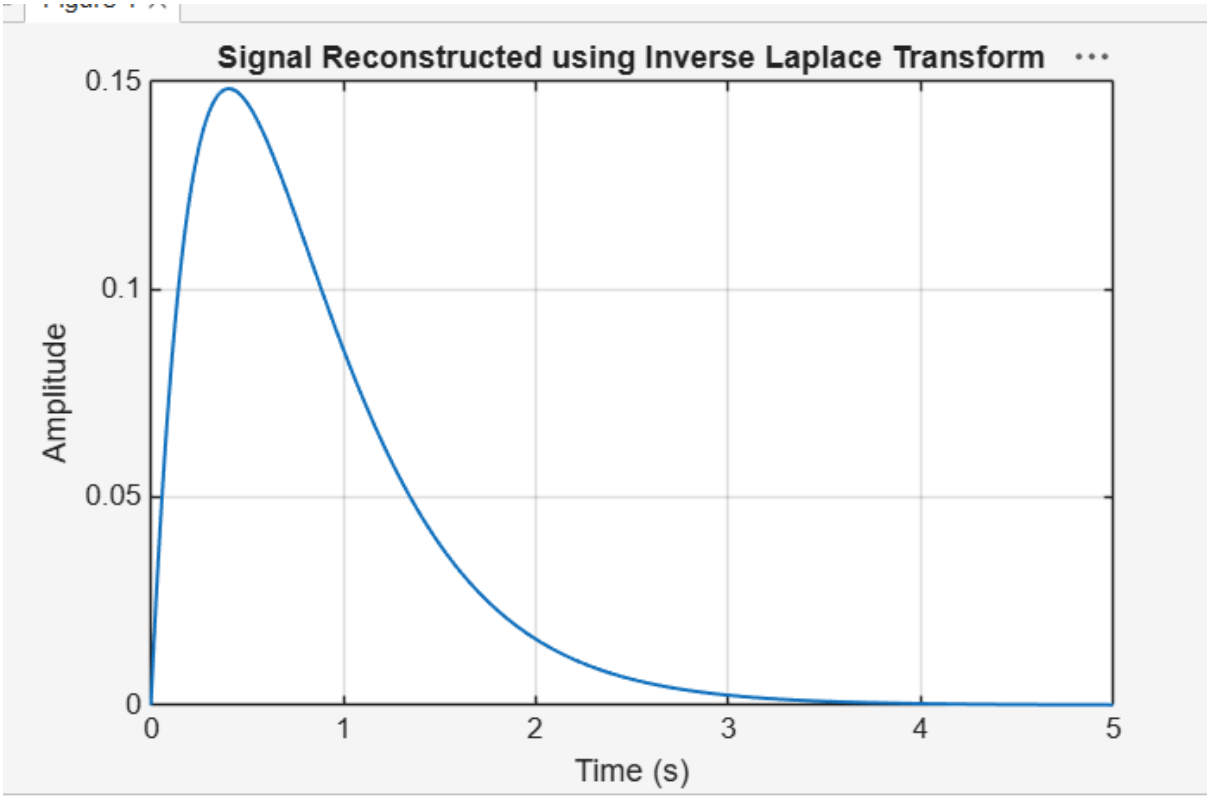
% Calculate inverse Laplace transform
x = ilaplace(X,s,t);
```

```
disp('Inverse Laplace Transform x(t) =');
pretty(x);

% Reconstruct the signal
tt = 0:0.001:5;
xx = double(subs(x,t,tt));

% Plot the reconstructed signal
figure;
plot(tt,xx,'LineWidth',1.2);
grid on;
xlabel('Time (s)');
ylabel('Amplitude');
title('Signal Reconstructed using Inverse Laplace Transform');
```

OUTPUT:



(iii) To determine the Region of Convergence (ROC) of Laplace transforms and analyze its significance in characterizing continuous-time signals and systems.

Software / Laboratory Environment:

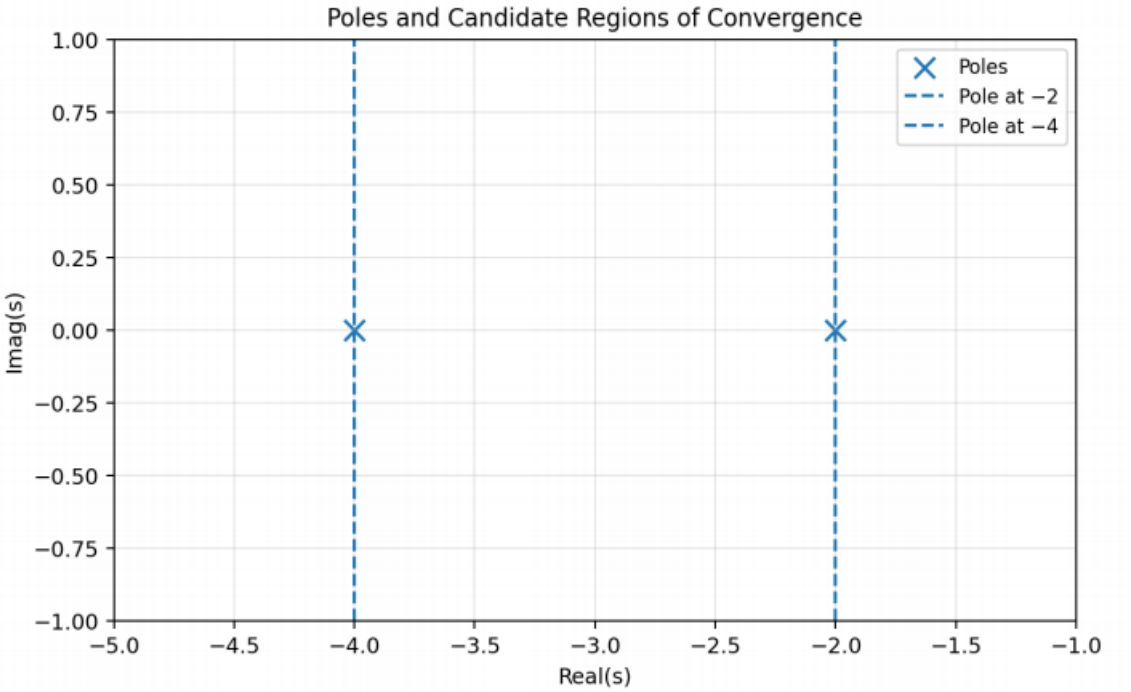
MATLAB

MATLAB Coding:

```
clc;
clear;
close all;b = [1 -0.5];
a = [1 -1.2 0.32];
z = roots(b);
p = roots(a)
disp('Zeros:');
disp(z);
```

```
disp('Poles:');
disp(p);
figure;
zplane(b,a);
grid on;
title('Pole-Zero Plot and Unit Circle');
hold on;
theta = linspace(0,2*pi,500);
plot(cos(theta),sin(theta),'k--','LineWidth',1.2);
legend('Zeros','Poles','Unit Circle');
fprintf('\nPole magnitudes:\n');
disp(abs(p));
if all(abs(p) < 1)
    fprintf('All poles are inside the unit circle.\n');
    fprintf('The causal system is stable.\n');
else
    fprintf('At least one pole is on or outside the unit circle.\n');
    fprintf('The causal system is unstable.\n');
end
```

OUTPUT:



RUBRICS:

Objective	Assessment Criterion	Excellent – 4	Good – 3	Satisfactory – 2	Needs Improvement – 1	Weightage	Marks(10)	Marks obtained
(ii) To determine the magnitude and phase spectra of continuous-time signals and	Fourier Transform Computation & MATLAB Implementation	Computes X(jω) of every given CT signal correctly, analytically and numerically .	Computes the spectra correctly, with minor errors in scaling, sampling rate, frequency axis or	Computes the spectra for only some signals, or has noticeable errors in code, scaling or	Unable to compute the Fourier transform or produce meaningful spectra .	20	2	

analyze their significant spectral components.		**Uses a suitable time range, sampling rate and frequency axis in rad/s or Hz.**	**parameters.**	**frequency axis, and needs assistance.**				
	Magnitude & Phase Spectrum Plots	**Plots $\|X(j\omega)\|$ and $\angle X(j\omega)$ correctly for every signal, with proper scaling, titles, axis labels and grid. Handles phase wrapping and phase near zero-magnit**	**Plots both spectra correctly, with minor errors in scaling, labelling, phase wrapping or axis range.**	**Plots the spectra with noticeable errors in scaling, phase handling or labelling, and needs guidance.**	**Unable to plot the magnitude and phase spectra.**	**20**	**2**	

		ude points correctly.						
	Spectral Component Analysis	**Correctly identifies dominant frequencies, peaks, bandwidth and the frequency range holding most of the signal energy. Uses symmetry (even magnitude, odd phase for real signals**	**Identifies most significant spectral components and bandwidth, with minor errors or limited justification.**	**Identifies only basic spectral components. Bandwidth or symmetry reasoning is incomplete, and needs guidance.**	**Unable to identify or justify the significant spectral components.**	**20**	**2**	

		) and relates the spectral lines or lobes to the signal parameters.						
	Analytical vs Numerical Verification	**Compares the analytical and MATLAB spectra, quantifies the difference and explains its causes such as truncation, sampling and spectral leakage.**	**Compares analytical and numerical results correctly, with brief explanation of differences.**	**Makes a basic comparison with noticeable errors, and the explanation of differences is incomplete.**	**Unable to verify the numerical spectra against the analytical results.**	**20**	**2**	

		Shows how signal parameters change the spectrum.						
	Engineering Analysis, Record & Interpretation	**Provides technically accurate interpretation of the results and relates them to the physical or engineering meaning of the signal. Maintains a complete, well-organi**	**Maintains a proper record with correct interpretation, with minor omissions in explanation or conclusions.**	**Record and interpretation are partial or only descriptive, and need guidance.**	**Incomplete record, limited or incorrect interpretation of the results.**	**20**	**2**	

		zed record with observations, tables, plots and conclusions.						
					Total	**100**	**10**	